



PAKISTAN STOCK EXCHANGE LIMITED

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Ref. No. PSX/ GEN - 313

To Whom It May Concern

28th Feb, 2024

Mr. Muhammad Aqib, an employee of Catalyst I.T Solutions was deputed to the Pakistan Stock Exchange (PSX) in the capacity of Software Developer in March 2022. In this position, Aqib has played a pivotal role in the design and development of a major Order Management System - the *Jade Trading Terminal (JTT)* and thereby extended significant contribution towards the success of one of the most noteworthy projects ever undertaken at the PSX, namely the implementation of the ***New Trading and Surveillance System***.

During his engagement in the project, Aqib delivered a number of highly critical components integral to the success of the *Jade Trading Terminal*:

Validation Server:

- Engineered a resilient validation server capable of managing thousands of client requests over TCP, ensuring seamless operations under high demand.
- Implemented stringent user validation mechanisms to safeguard secure access for trading activities within the Stock Market, bolstering overall system integrity and reliability.

MBS (Market by Symbol):

- Established robust connections with the Market Data Gateway (MDGW) utilizing the FIX Protocol for seamless reception and processing of tick-by-tick market feeds.
- Engineered an efficient in-memory mapping system for symbols, facilitating real-time access to critical data on the top 30 depths of buy and sell queued orders, seamlessly delivered to clients through TIBCO message queues.

Exchange-Traded Fund (ETF Circuit Breaker):

- Developed a server to calculate the upper and lower price limits of scripts based on the current KIBOR rate and DTM of underlying symbols, prior to the pre-opening of the next trading day.

CDC Interface:

- Developed RESTful APIs deployed in the PSX environment, exclusively accessible to the Central Depository Company (CDC), facilitating the seamless settlement of shares into the PSX database at the end of each trading day.

Karachi Internet Trading System (KiTS):

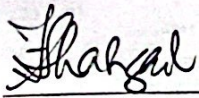
- Enhanced the online trading system to align seamlessly with the requirements of the new Trading System (JTT), ensuring optimal compatibility and functionality.
- Implemented diverse new markets and executed meticulous risk management strategies as specified by PSX, guaranteeing adherence to regulatory standards and operational excellence.

Aqib has consistently demonstrated exceptional technical expertise, dedication, and professionalism throughout his engagement with the Pakistan Stock Exchange. His invaluable contributions have been instrumental in the successful development and implementation of critical components of the Jade Trading Terminal and his constant support is greatly appreciated by the PSX IT team.

It may be said with certainty that Aqib will be an asset for any organization and project group and his skills and experience will continue to yield success in his future endeavors.

Sincerely,

Pakistan Stock Exchange



DGM - IT



CIO